

```
acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180$ scp lab04_v3.c
table3.sh cpdelosreyes@overqueen/home/cpdelosreyes
cpdelosreyes@overqueen/home/cpdelosreyes: No such file or directory
```

```
acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180$ scp lab04_v3.c
table3.sh cpdelosreyes@10.0.9.20:~
cpdelosreyes@10.0.9.20's password:
Connection closed by 10.0.9.20 port 22
scp: Connection closed
```

```
acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180$ scp lab04_v3.c
table3.sh cpdelosreyes@10.0.9.20:~
cpdelosreyes@10.0.9.20's password:
lab04_v3.c
100% 13KB 1.8MB/s 00:00
table3.sh
100% 6749 685.2KB/s 00:00
acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180$ ssh
cpdelosreyes@10.0.9.125
The authenticity of host '10.0.9.125 (10.0.9.125)' can't be established.
ED25519 key fingerprint is SHA256:vl7W8nM3tEM5su+LVjapkA7f9DuER2O7CMlrEgLOjLE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.9.125' (ED25519) to the list of known hosts.
cpdelosreyes@10.0.9.125's password:
cpdelosreyes@fanny:~$ cd ~
cpdelosreyes@fanny:~$ gcc -o lab04_v3 lab04_v3.c -lpthread -lm
cpdelosreyes@fanny:~$ chmod +x lab04_v3 table3.sh
cpdelosreyes@fanny:~$ ssh-keygen -t rsa -N ""
Generating public/private rsa key pair.
Enter file in which to save the key (/home/cpdelosreyes/.ssh/id_rsa):
Created directory '/home/cpdelosreyes/.ssh'.
Your identification has been saved in /home/cpdelosreyes/.ssh/id_rsa
Your public key has been saved in /home/cpdelosreyes/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:bwiFiFd4DnoYbv81VkxE8bfEsKFtwpYiODSZy5n+Zx8
cpdelosreyes@fanny.swarm.ics.uplb.edu.ph
The key's randomart image is:
+---[RSA 3072]----+
|  o+. o+.o  |
| .o==.. ..= = |
| ..==Bo oo* = + |
| =.*..o ooo o . |
| . + . S. . . |
```

```
| o .+o |
| o o..E |
| o o. . |
| o .. |
```

+----[SHA256]-----+

cpdelosreyes@fanny:~\$ ssh-copy-id cpdelosreyes@10.0.9.168

/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed:

"/home/cpdelosreyes/.ssh/id_rsa.pub"

The authenticity of host '10.0.9.168 (10.0.9.168)' can't be established.

ED25519 key fingerprint is SHA256:vl7W8nM3tEM5su+LVjapkA7f9DuER2O7CMIrEgLOjLE.

This key is not known by any other names.

Are you sure you want to continue connecting (yes/no/[fingerprint])?

/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed

The authenticity of host '10.0.9.168 (10.0.9.168)' can't be established.

ED25519 key fingerprint is SHA256:vl7W8nM3tEM5su+LVjapkA7f9DuER2O7CMIrEgLOjLE.

This key is not known by any other names.

Are you sure you want to continue connecting (yes/no/[fingerprint])? yes

/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys

cpdelosreyes@10.0.9.168's password:

5f6b8e5a151da65450d3ec559b0511cc

cpdelosreyes@fanny:~\$ ssh-copy-id cpdelosreyes@10.0.9.168

/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed:

"/home/cpdelosreyes/.ssh/id_rsa.pub"

The authenticity of host '10.0.9.168 (10.0.9.168)' can't be established.

ED25519 key fingerprint is SHA256:vl7W8nM3tEM5su+LVjapkA7f9DuER2O7CMIrEgLOjLE.

This key is not known by any other names.

Are you sure you want to continue connecting (yes/no/[fingerprint])?

/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed

The authenticity of host '10.0.9.168 (10.0.9.168)' can't be established.

ED25519 key fingerprint is SHA256:vl7W8nM3tEM5su+LVjapkA7f9DuER2O7CMIrEgLOjLE.

This key is not known by any other names.

Are you sure you want to continue connecting (yes/no/[fingerprint])? yes

/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys

cpdelosreyes@10.0.9.168's password:

5f6b8e5a151da65450d3ec559b0511cc

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'cpdelosreyes@10.0.9.168'"

and check to make sure that only the key(s) you wanted were added.

```
cpdelosreyes@fanny:~$  
cpdelosreyes@fanny:~$ 5f6b8e5a151da65450d3ec559b0511cc  
-bash: 5f6b8e5a151da65450d3ec559b0511cc: command not found  
cpdelosreyes@fanny:~$ ssh cpdelosreyes@10.0.9.168 "echo hello"  
hello  
cpdelosreyes@fanny:~$ ssh cpdelosreyes@10.0.9.158 "echo hello"  
The authenticity of host '10.0.9.158 (10.0.9.158)' can't be established.  
ED25519 key fingerprint is SHA256:vl7W8nM3tEM5su+LVjapkA7f9DuER2O7CMIrEgLOjLE.  
This host key is known by the following other names/addresses:  
  ~/.ssh/known_hosts:1: 10.0.9.168  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '10.0.9.158' (ED25519) to the list of known hosts.  
hello  
cpdelosreyes@fanny:~$
```

LAST RUN PROBLEM

```
cpdelosreyes@fanny:~$ ssh cpdelosreyes@10.0.9.158 "echo hello"  
The authenticity of host '10.0.9.158 (10.0.9.158)' can't be established.  
ED25519 key fingerprint is SHA256:vl7W8nM3tEM5su+LVjapkA7f9DuER2O7CMIrEgLOjLE.  
This host key is known by the following other names/addresses:  
  ~/.ssh/known_hosts:1: 10.0.9.168  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '10.0.9.158' (ED25519) to the list of known hosts.  
hello  
cpdelosreyes@fanny:~$ ssh cpdelosreyes@10.0.9.125  
The authenticity of host '10.0.9.125 (10.0.9.125)' can't be established.  
ED25519 key fingerprint is SHA256:vl7W8nM3tEM5su+LVjapkA7f9DuER2O7CMIrEgLOjLE.  
This host key is known by the following other names/addresses:  
  ~/.ssh/known_hosts:1: 10.0.9.168  
  ~/.ssh/known_hosts:4: 10.0.9.158  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '10.0.9.125' (ED25519) to the list of known hosts.  
Last login: Thu Apr 16 15:27:12 2026 from 10.0.4.69  
cpdelosreyes@fanny:~$ cd ~  
cpdelosreyes@fanny:~$ ./table3.sh  
Error: /home/cadelosreyes/lab04_v3 not found on this drone.  
Upload to overqueen first: scp lab04_v3.c cadelosreyes@10.0.9.19:~/  
Then compile: gcc -o lab04_v3 lab04_v3.c -lpthread -lm  
cpdelosreyes@fanny:~$
```

fanny.swarm.ics.uplb.edu.ph	10.0.9.125
--	------------

27	sora.swarm.ics.uplb.edu.ph	10.0.9.168
28	natan.swarm.ics.uplb.edu.ph	10.0.9.158
29	rafaela.swarm.ics.uplb.edu.ph	10.0.9.166

30	arlott.swarm.ics.uplb.edu.ph	10.0.9.117
31	alucard.swarm.ics.uplb.edu.ph	10.0.9.176
32	mathilda.swarm.ics.uplb.edu.ph	10.0.9.156
33	dyrroth.swarm.ics.uplb.edu.ph	10.0.9.111
34	layla.swarm.ics.uplb.edu.ph	10.0.9.143
35	beatrix.swarm.ics.uplb.edu.ph	10.0.9.179
36	hilda.swarm.ics.uplb.edu.ph	10.0.9.136
37	yve.swarm.ics.uplb.edu.ph	10.0.9.173
38	selena.swarm.ics.uplb.edu.ph	10.0.9.167
39	leomord.swarm.ics.uplb.edu.ph	10.0.9.109
40	phoveus.swarm.ics.uplb.edu.ph	10.0.9.164
41	harith.swarm.ics.uplb.edu.ph	10.0.9.133
42	khaleed.swarm.ics.uplb.edu.ph	10.0.9.186

MD5 Hash Generator

 Add to Fav

New

Save & Share

Enter the plain or Cipher Text:

Sample ↺📁💾🗑️📄

cpdelosreyes202315797

Size : 21 B, 21 Characters

☒ Auto

Generate

File..

Load URL

Result of MD5 Generated Hash:

Upper CaseLower Case📄

5f6b8e5a151da65450d3ec559b0511cc

Generated Hash:

5f6b8e5a151da65450d3ec559b0511cc

Table 1:

=====

Table 1 complete! Results saved to: lab04_table1.csv

=====

n	t	Run 1	Run 2	Run 3	Average
4000	2	0.014540	0.018172	0.018506	.017072
4000	4	0.022927	0.022501	0.023492	.022973
4000	8	0.027694	0.027366	0.033009	.029356
4000	16	0.038022	0.033686	0.033495	.035067
8000	2	0.056518	0.060112	0.055731	.057453
8000	4	0.085382	0.087451	0.086881	.086571

8000	8	0.102653	0.099831	0.111726	.104736
8000	16	0.124436	0.122716	0.126551	.124567
16000	2	0.204243	0.199779	0.208155	.204059
16000	4	0.307341	0.296739	0.303263	.302447
16000	8	0.414202	0.394432	0.398659	.402431
16000	16	0.433494	0.459253	0.458044	.450263

Table 2:

=====

Table 2 complete! Results saved to: lab04_table2.csv

=====

n	t	Run 1	Run 2	Run 3	Average
4000	2	0.016576	0.015553	0.015187	.015772
4000	4	0.023021	0.024071	0.024345	.023812
4000	8	0.029421	0.030629	0.029055	.029701
4000	16	0.035423	0.034414	0.035923	.035253
8000	2	0.052038	0.056003	0.056865	.054968
8000	4	0.088928	0.086846	0.085981	.087251
8000	8	0.103423	0.101003	0.109725	.104717
8000	16	0.130102	0.122868	0.128082	.127017
16000	2	0.219621	0.215163	0.225091	.219958
16000	4	0.343269	0.333158	0.332469	.336298
16000	8	0.414651	0.385119	0.414849	.404873
16000	16	0.461068	0.494783	0.489821	.481890


```
#!/bin/bash

# =====
# Table 3: Different PCs (ICS Swarm)
# =====
# Run this on the MASTER drone. It will SSH into slave drones
# to start slaves remotely, then run the master locally.
#
# BEFORE RUNNING:
#   1. Edit DRONE_IPS below with actual drone IPs
#   2. Make sure lab04_v3 is compiled and accessible on all
#       drones (upload to overqueen, drones share filesystem)
#   3. Make sure ssh key-based auth works (or you'll be
#       prompted for password for every slave on every run)
#   4. Run this script ON one of the drones (NOT overqueen)
#
# Usage:
#   chmod +x table3.sh
#   ./table3.sh
#
# Output:
#   - lab04_table3.csv (raw timing data)
#   - formatted table printed to terminal
# =====

# =====
# EDIT THESE BEFORE RUNNING
# =====

# your swarm username (UP email without @up.edu.ph)
USERNAME="cpdelosreyes"

# path to lab04_v3 on the swarm (shared filesystem from overqueen)
# adjust this if your home directory path is different
REMOTE_EXE="/home/$USERNAME/lab04_v3"

# the drone you are running this script on (master)
MASTER_IP="10.0.9.125"

# all available drone IPs for slaves (list at least 16)
# the script will pick the first t drones from this list
# IMPORTANT: IPs are dynamic! Check http://10.0.9.19/ for today's IPs
DRONE_IPS=(
```

```
"10.0.9.168"
"10.0.9.158"
"10.0.9.166"
"10.0.9.117"
"10.0.9.176"
"10.0.9.156"
"10.0.9.111"
"10.0.9.143"
"10.0.9.179"
"10.0.9.136"
"10.0.9.173"
"10.0.9.167"
"10.0.9.109"
"10.0.9.164"
"10.0.9.133"
"10.0.9.186"
)

# =====
# DO NOT EDIT BELOW THIS LINE
# =====

EXE="$REMOTE_EXE"
MASTER_PORT=5000
SLAVE_PORT=5001
CSV="lab04_table3.csv"

N_VALUES=(4000 8000 16000)
T_VALUES=(2 4 8 16)
RUNS=3

# ssh options: skip host key check, timeout after 5 seconds
SSH_OPTS="-o StrictHostKeyChecking=no -o ConnectTimeout=5"

# check if executable exists locally
if [ ! -f "$EXE" ]; then
    echo "Error: $EXE not found on this drone."
    echo "Upload to overqueen first: scp lab04_v3.c $USERNAME@10.0.9.20:~"
    echo "Then compile: gcc -o lab04_v3 lab04_v3.c -lpthread -lm"
    exit 1
fi
```

```

# check we have enough drones
MAX_T=16
if [ "${#DRONE_IPS[@]}" -lt "$MAX_T" ]; then
    echo "Warning: Only ${#DRONE_IPS[@]} drone IPs listed. t=16 needs 16
drones."
    echo "For t values that need more drones than available, slaves will"
    echo "share drones using different ports."
    echo ""
fi

echo "====="
echo "Table 3: Different PCs (ICS Swarm)"
echo "====="
echo "Master drone: $MASTER_IP"
echo "Slave drones: ${DRONE_IPS[*]:0:4} ..."
echo "Username: $USERNAME"
echo ""

# csv header
echo "n,t,run,time" > "$CSV"

TOTAL=$(( ${#N_VALUES[@]} * ${#T_VALUES[@]} * RUNS ))
COUNT=0

for n in "${N_VALUES[@]"; do
    for t in "${T_VALUES[@]"; do

        # generate config file
        CONFIG="config_swarm_t${t}.txt"
        echo "master $MASTER_IP $MASTER_PORT" > "$CONFIG"

        # assign drones to slaves
        # if enough drones: 1 slave per drone, all use same port
        # if not enough: wrap around drones, use different ports
        SLAVE_IPS=()
        SLAVE_PORTS=()
        DRONE_COUNT=${#DRONE_IPS[@]}

        for ((i=0; i<t; i++)); do
            DRONE_IDX=$((i % DRONE_COUNT))
            DRONE_ROUND=$((i / DRONE_COUNT))
            S_IP="${DRONE_IPS[$DRONE_IDX]}"

```

```

        S_PORT=$((SLAVE_PORT + DRONE_ROUND))

        SLAVE_IPS+=("$S_IP")
        SLAVE_PORTS+=("$S_PORT")
        echo "slave $S_IP $S_PORT" >> "$CONFIG"
    done

    for ((run=1; run<=RUNS; run++)); do
        COUNT=$((COUNT + 1))
        printf "[%d/%d] n=%d, t=%d, run %d ... " "$COUNT" "$TOTAL"
"$n" "$t" "$run"

        # copy config to shared filesystem so drones can access it
        cp "$CONFIG" "$HOME/$CONFIG" 2>/dev/null

        # start slaves on remote drones via ssh
        SLAVE_PIDS=()
        for ((i=0; i<t; i++)); do
            S_IP="${SLAVE_IPS[$i]}"
            S_PORT="${SLAVE_PORTS[$i]}"

            # ssh into drone, run slave in background, redirect output
            ssh $SSH_OPTS "$USERNAME@$S_IP" \
                "cd ~ && $EXE $n $S_PORT 1 $CONFIG > /dev/null 2>&1" &
            SLAVE_PIDS+=($!)
        done

        # wait for slaves to start listening on remote drones
        sleep 3

        # run master locally and capture output
        MASTER_OUTPUT=$((EXE $n $MASTER_PORT 0 "$CONFIG" 2>&1)

        # extract time from master output
        TIME_VAL=$(echo "$MASTER_OUTPUT" | grep -oP 'time
elapsed:\s+\K[\d.]+')

        if [ -n "$TIME_VAL" ]; then
            printf "%s seconds\n" "$TIME_VAL"
            echo "$n,$t,$run,$TIME_VAL" >> "$CSV"
        else
            printf "ERROR\n"

```

```

        echo "$n,$t,$run,ERROR" >> "$CSV"
        # print full output for debugging
        echo "    Debug output:"
        echo "$MASTER_OUTPUT" | head -5 | sed 's/^/    /'
    fi

    # wait for ssh sessions to finish
    for pid in "${SLAVE_PIDS[@]}; do
        wait "$pid" 2>/dev/null
    done

    # brief pause to release ports
    sleep 2
done

# clean up temp config
rm -f "$CONFIG"
rm -f "$HOME/$CONFIG" 2>/dev/null
done
done

echo ""
echo "====="
echo "Table 3 complete! Results saved to: $CSV"
echo "====="
echo ""

# print formatted table
printf "%-10s %-5s %-14s %-14s %-14s %-14s\n" "n" "t" "Run 1" "Run 2" "Run
3" "Average"
printf "%s\n"
"-----"

for n in "${N_VALUES[@]}; do
    for t in "${T_VALUES[@]}; do
        TIMES=()
        for ((run=1; run<=RUNS; run++)); do
            val=$(grep "^${n},${t},${run}," "$CSV" | cut -d',' -f4)
            TIMES+=("$val")
        done

```

```
        if [ "${#TIMES[@]}" -eq "$RUNS" ] && [ -n "${TIMES[0]}" ] && [
"${TIMES[0]}" != "ERROR" ]; then
            AVG=$(echo "scale=6; (${TIMES[0]} + ${TIMES[1]} + ${TIMES[2]})
/ 3" | bc)
            printf "%-10s %-5s %-14s %-14s %-14s %-14s\n" "$n" "$t"
"${TIMES[0]}" "${TIMES[1]}" "${TIMES[2]}" "$AVG"
        else
            printf "%-10s %-5s %-14s %-14s %-14s %-14s\n" "$n" "$t"
"ERROR" "ERROR" "ERROR" "ERROR"
        fi
    done
    echo ""
done
```